

Anisur Rahman

Sr. Software Engineer at AppsCode

✉ sunny.cse7575@gmail.com ☎ +88 01540179777 🌐 GitHub 🔗 LinkedIn 📧 Medium

RESEARCH INTERESTS

Cloud Computing, Database Internals, Distributed Systems, Cloud-Native, Kubernetes, Container Orchestration, Storage Systems, Replication, Erasure Coding, Compression, Algorithm Design, Software Engineering, DevOps

EDUCATION

Daffodil International University

BSc in Computer Science and Engineering (CGPA: 3.87/4.00)

Dhaka, Bangladesh

Jan 2019 - Dec 2022

TECHNICAL SKILLS

Languages: Go (Golang), C++, Bash

Platforms: Linux, Docker, Kubernetes, OpenShift, Cloud-Native, AWS, GCP

Technologies: REST, gRPC, Protocol Buffers, Makefile, Helm, CI/CD, GitOps, FluxCD, Prometheus, Grafana

Storage systems: LVM, ZFS, K8s-CSI, Restic, Wal-G, Velero, Rook-Ceph, PostgreSQL, MySQL

Core Strengths: Microservice, Distributed Systems, Raft, Concurrency, Parallel Processing, Software Architecture

ACCOMPLISHMENTS

Publications

- *OPTiMiZiNG CONVOLUTIONAL NEURAL NETWORKS FOR REAL-TiME FABRiC DEFECT DETECTiON*
- Optimized Convolutional Neural Networks (CNNs) using advanced mathematical formulations [[paper](#)].
- Model accuracy of 99.39% — the highest recorded for any existing model.

Opensource Projects

Contributor to WAL-G, Stash, Go-Shell, and IBM MultiCloud Gateway.

- Multi-writer support for the go-shell. [[PR](#)] • External DB secret reconciliation for IBM MCG. [[PR](#)]
- HookExecutionPolicy. [[PR](#)] • Charset converter to non-UTF-8. [[PR](#)] • TLS Verification support. [[PR](#)]

Competitive Programming

2000+ problems solved across Codeforces, Atcoder, UVA, Spoj, etc.

- CodeChef Master (Max Rating: [2195](#)), Codeforces Expert ([1639](#))
- Rank 3.1%: ICPC, Dhaka Regional 2022. [[DIU-Sedulous](#)], Rank 3.4%: BUET IUPC Contest 2022. [[Standings](#)]

WORK EXPERIENCE

Senior Software Engineer

AppsCode Ltd, Dhaka, Bangladesh

January 2023 - Present

- **Golang:** Write high-performance Go code for distributed, cloud-native applications.
- **DBaaS:** Developed k8s-native managed database and disaster recovery platforms ([KubeDB](#), [KubeStash](#)) ensuring 99.99% uptime through automated failover for production clusters across global enterprises.
- **Parallel Task Scheduling System:** Built a distributed scheduler for parallel DB backup and restore, delivering 30% faster performance.
- **Multi-Cloud Gateway:** Integrated [NooBaa](#) Multi-Cloud Gateway into KubeStash, enabling cross-cloud S3 bucket replication to resolve storage failover.
- **CSI driver:** Extended the Topo-LVM CSI driver to support snapshot chunking and offloading to public object storage, ensure resilience during k8s nodes failure for hybrid and on-prem setups.
- **DB Migrator:** Architected and developed a database migration framework for *RDS* ⇌ DBaaS in k8s (PG and MySQL), optimized with parallel processing and snapshot dumping, handling multi-terabyte datasets.
- **SRE:** Ensured reliability of production Kubernetes clusters on bare metal using distributed storage (Rook-Ceph), resolved critical production issues for Orange, Nokia, Huma, and Emerson, ensuring 99.9% uptime.
- **Presentation:** Authored webinars, [documentation](#), and [blogs](#) about DBaaS management in k8s. [[1](#)], [[2](#)], [[3](#)], [[4](#)]

PROJECTS

KubeStash

Go • Kubernetes • Helm • Kubebuilder • Cloud • Concurrency • Restic • Prometheus • Grafana

A complete disaster recovery solution for Databases, Workload applications and Cluster for Kubernetes environment.

- Led development and ownership for 2.5 years from system design to release.
- Added features (retry, validation, and failure recovery) to make it a leading k8s disaster recovery solution.
- Added bucket as backup storage for all major clouds with cloud identity integration, e.g., IRSA, Pod identity.
- Architected sharded operators and controllers for large-scale, parallel backups across approx 1000+ DBaaS.
- Engineered high-performance backup & restore plugins (concurrent, snapshot, PITR) for 10+ databases.
- Wrote Prometheus exporters and designed a Grafana monitoring dashboard.
- Authored automated E2E and chaos tests to validate reliability under 1000 database workloads.

KubeDB WAL-G

Go • gRPC • Databases • Distributed System • Chunking

Wal Archival and Restoration for Databases in the Cloud

- Built a WAL archival and restoration framework supporting Point-in-Time Recovery (PITR) for KubeDB databases.
- Implemented continuous WAL shipping to any object storage.
- Architected a gRPC proxy server for WAL streaming in Kubernetes distributed environments.
- Automated DBaaS failover detection and restore from WAL backup.

PROJECTS

TopoLvm CSI Driver

Go • gRPC • Protocol Buffers • LVM • Chunking • Restic • Prometheus • Grafana

Restic-based Volumes snapshot feature for kubernetes PVC

- Architected a gRPC server to take volume snapshot by using Kubernetes CSI workflows.
- Efficient backup and restore operations for persistent volumes for hybrid and on-prem setups.
- Wrote Prometheus exporters for monitoring volume snapshot operations.

Migrator

Go • gRPC • Protocol Buffers • Prometheus

High-performance data migration engine for PostgreSQL & MySQL

- Architect and Developed parallel/concurrent pipelines for high-volume database data transfer.
- Designed a gRPC-based control plane for managing complex migration workflows.
- Achieved approx +-10 hours to migrate 2 TB of MySQL data from RDS to kubedb managed DBaaS.

TaskQueue

Go • Concurrency • Parallel Processing

A Kubernetes controller that manages a queue of distributed task in kubernetes environment

- Built a Kubernetes controller for managing distributed task queues with concurrent processing capabilities.
- Implemented worker pool pattern for efficient task execution and resource management.